

Milestone 2 Report

MedAssistAI

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July 24, 2026

Contents

1	Project Description	2
2	Dataset Used	2
3	Data Preprocessing	2
4	Training Pipeline & Model Details	2
5	Results & Evaluation	3

1 Project Description

This milestone focuses on building, optimizing, and evaluating a robust multi-model machine learning pipeline designed to classify medical conditions across hundreds of distinct disease classes based on symptom features. The workflow includes extensive data preprocessing, the training of five distinct tree-based models, and the creation of highly accurate voting ensembles.

2 Dataset Used

The project utilizes the `Final_Augmented_dataset_Diseases_and_Symptoms.csv` dataset, which was loaded directly from a mounted Google Drive environment.

- The raw dataset initially contained 246,945 rows and 378 feature columns.

3 Data Preprocessing

The raw data underwent rigorous cleaning to standardize the inputs and reduce computational overhead:

- **Memory Optimization:** All feature columns were cast to the `np.uint8` format to optimize system memory footprint.
- **Feature Selection:** Zero-variance features (columns with one or fewer unique values) were identified and dropped.
- **Deduplication:** Exact duplicate rows were scrubbed from the dataset.
- **Rare Class Filtering:** Disease classes containing fewer than 2 samples were removed to ensure viable training distributions.
- **Label Encoding:** Target diseases were encoded using `LabelEncoder`, resulting in a final count of 727 valid target classes.
- **Data Splitting:** The cleaned dataset, comprising 189,601 rows and 328 features, was divided via a stratified split into an 80% training set (151,680 rows) and a 20% testing set (37,921 rows).

4 Training Pipeline & Model Details

Five distinct baseline models were trained on the processed data. A critical evaluation fix (`expand_proba`) was implemented to remap Scikit-Learn-style model probabilities into the full 727 class-ID space, ensuring column indices strictly equaled true class IDs during predictions.

- **XGBoost:**
 - Trained with GPU acceleration (`device: "cuda"`) using the histogram tree method (`tree_method: "hist"`).

- Hyperparameters included a `max_depth` of 6, learning rate (`eta`) of 0.05, `subsample` of 0.8, and `colsample_bytree` of 0.7.
- Configured for 1,000 boosting rounds with early stopping at 50 rounds; the model achieved its best iteration at round 92.
- **LightGBM:**
 - Features were renamed to generic, collision-free names (e.g., `feature_0`) to eliminate JSON/bracket formatting errors during training.
 - Hyperparameters included 300 estimators, a `max_depth` of 8, 64 `num_leaves`, and a learning rate of 0.1.
- **Gradient Boosting (`HistGradientBoostingClassifier`):**
 - Optimized for training speed.
 - Configured with a maximum of 30 iterations, a `max_depth` of 5, and a learning rate of 0.2.
- **Random Forest:**
 - Configured to be memory-safe but strong by bootstrapping only 50% of the rows per tree (`max_samples=0.5`).
 - Hyperparameters included 120 estimators and a `max_depth` of 18.
- **Decision Tree:**
 - Trained with a `max_depth` of 25, which yielded 340 leaves.

5 Results & Evaluation

The models were benchmarked on Top-1, Top-3, and Top-5 accuracy, alongside Weighted F1 scores. Two ensembles were tested: an equal-weight “All 5” ensemble, and a Weighted Ensemble where models were assigned weights based on a composite average of their Top-1, Top-3, and Top-5 accuracies.

As detailed in Table 1, **XGBoost** emerged as the strongest individual baseline model, achieving the highest Top-1 accuracy of 79.92% and a Weighted F1 score of 0.7905. However, when evaluating the models’ ability to place the correct diagnosis within the top few candidates, the **Weighted Voting Ensemble** demonstrated superior performance. By weighting each model’s contribution based on its composite accuracy, the ensemble achieved the highest overall Top-3 (92.53%) and Top-5 (96.13%) accuracies. The unweighted “All 5” Ensemble also performed exceptionally well, trailing just behind the weighted version.

Conversely, the Decision Tree and LightGBM models underperformed relative to the XGBoost and Gradient Boosting algorithms. This performance gap highlights the necessity of advanced gradient boosting techniques and weighted ensembles for handling highly sparse, 727-class medical datasets.

Table 1: Model Comparison Benchmark

Model	Top-1	Top-3	Top-5	Weighted F1
XGBoost	0.7992	0.9204	0.9483	0.7905
Voting Ensemble (Weighted)	0.7804	0.9253	0.9613	0.7899
Voting Ensemble (All 5)	0.7729	0.9246	0.9609	0.7851
Gradient Boosting	0.7531	0.9001	0.9372	0.7502
Random Forest	0.6444	0.7562	0.7872	0.6488
LightGBM	0.5293	0.8092	0.8727	0.5848
Decision Tree	0.1481	0.1806	0.2037	0.1822

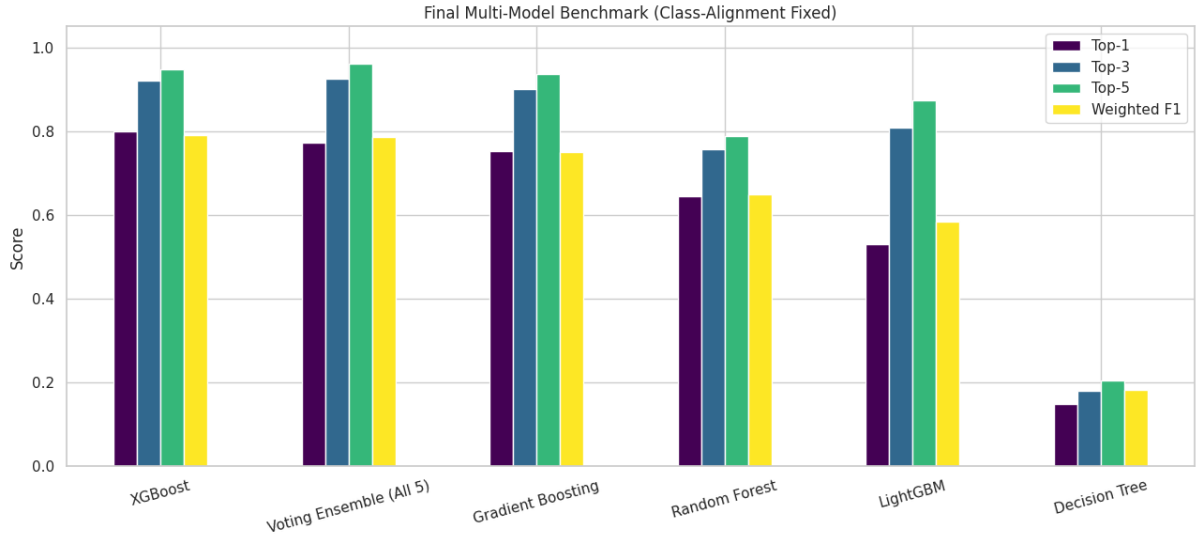


Figure 1: Final Multi-Model Benchmark (Equal vs Weighted Voting)